Section B: API Endpoint Plan

The API Endpoint Plan defines the RESTful endpoints that will be implemented in Part 2 of the RaceDay system. It outlines the HTTP methods, routes, purpose of each endpoint, required user role, request body, and expected responses (Microsoft, 2026b). The endpoints are designed to support authentication, user profiles, event and category management, participant enrolments, results, and image uploads while enforcing role-based access

Authentication

These are public because the user isn't authenticated yet

Method	Route	Description	Role	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as an Organizer or Participant. Password is hashed before storage.	Public	firstName, lastName, email, password, phoneNumber, role	201 Created – account created. 400 Bad Request – invalid data. 409 Conflict – email already exists.
POST	/api/auth/login	Authenticates a registered user and establishes a session containing their user ID and role.	Public	email, password	200 OK – login successful. 401 Unauthorized – invalid credentials.
POST	/api/auth/logout	Ends the authenticated user's session.	Any authenticated user	None	200 OK – session ended. 401 Unauthorized – not authenticated.

User Profile

Both roles need to be able to view/update their own information

Method	Route	Description	Role	Request Body	Expected Response
GET	/api/users/profile	Returns the profile information of the currently authenticated user.	Any authenticated user	None	200 OK – user profile. 401 Unauthorized – not logged in.
PUT	/api/users/profile	Updates the currently authenticated user's personal/profile information.	Any authenticated user	Updated profile fields	200 OK – updated profile. 400 Bad Request – invalid data. 401 Unauthorized – not logged in.

Events

This is the main Organizer functionality

Method	Route	Description	Role	Request Body	Expected Response
GET	/api/events	Returns upcoming RaceDay events.	Public	None / optional filters	200 OK – list of events.
GET	/api/events/{eventId}	Returns full information for a specific event.	Public	None	200 OK – event details. 404 Not Found – event does not exist.
POST	/api/events	Creates a new event.	Organizer	name, description, date, location, distance, eventType	201 Created – event created. 400 Bad Request – invalid data. 401 Unauthorized – not logged in. 403 Forbidden – Participant.

PUT	/api/events/{eventId}	Updates an event created by the logged-in Organizer.	Organizer	Updated event fields	200 OK – event updated. 404 Not Found – event does not exist. 403 Forbidden – user is not authorized to modify it.
DELETE	/api/events/{eventId}	Deletes an event managed by the logged-in Organizer.	Organizer	None	204 No Content – deleted. 404 Not Found – event does not exist. 403 Forbidden – not the event's Organizer.

Categories

Method	Route	Description	Role	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Returns all categories available for an event.	Public	None	200 OK – categories. 404 Not Found – event does not exist.
GET	/api/categories/{categoryId}	Returns details of a specific category.	Public	None	200 OK – category. 404 Not Found – category does not exist.
POST	/api/events/{eventId}/categories	Creates a category for an event.	Organizer	name, distance, ageGroup, price	201 Created – category created.

					400 Bad Request – invalid data. 403 Forbidden – not authorized.
PUT	/api/categories/{categoryId}	Updates an event category.	Organizer	Updated category fields	200 OK – category updated. 404 Not Found – category not found. 403 Forbidden – not authorized.
DELETE	/api/categories/{categoryId}	Deletes an event category.	Organizer	None	204 No Content – category deleted. 404 Not Found – category not found. 403 Forbidden – not authorized.

Event Enrolments

Method	Route	Description	Role	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enrolls the logged-in Participant in an event using their selected category.	Participant	categoryId	201 Created – enrolment recorded. 400 Bad Request – invalid category. 401

					Unauthorized – not logged in. 403 Forbidden – Organizer. 409 Conflict – already enrolled.
GET	/api/users/me/enrolments	Returns all events the logged-in Participant has enrolled in.	Participant	None	200 OK – participant's enrolments.
GET	/api/events/{eventId}/enrolments	Returns all Participants enrolled in a specific event.	Organizer	None	200 OK – enrolment list. 403 Forbidden – not the event's Organiser. 404 Not Found – event not found.
GET	/api/enrolments/{enrolmentId}	Returns details of a specific enrolment.	Authenticate user	None	200 OK – enrolment details. 403 Forbidden – user does not have access. 404 Not Found – enrolment not found.
PUT	/api/enrolments/{enrolmentId}	Updates an enrolment'	Organizer	status	200 OK – enrolment updated.

		s status where permitted.			403 Forbidden – not authorized. 404 Not Found – enrolment not found.

Results

Organizers capture results. Participants view their own results.

Method	Route	Description	Role	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Records a Participant's finish time and finishing position after an event.	Organizer	finishTime, finishingPosition	201 Created – result recorded. 400 Bad Request – invalid result. 403 Forbidden – not the event's Organiser. 404 Not Found – enrolment not found.
PUT	/api/results/{resultId}	Updates a recorded Participant result.	Organizer	finishTime, finishingPosition	200 OK – result updated. 403 Forbidden – not authorized. 404

					Not Found – result not found.
GET	/api/users/me/results	Returns the logged-in Participant's personal race history.	Participant	None	200 OK – personal results.
GET	/api/results/{resultId}	Returns details of a specific result.	Authenticated user	None	200 OK – result details. 403 Forbidden – user does not have access. 404 Not Found – result not found.

Event Banner Image

Method	Route	Description	Role	Request Body	Expected Response
POST	/api/events/{eventId}/image	Uploads or stores the banner image for an event using Azure Blob Storage.	Organizer	Multipart/form-data image file	201 Created – image uploaded and associated with event. 400 Bad Request – invalid file. 403 Forbidden – not event

					Organiser. 404 Not Found – event not found.
DELETE	/api/events/{eventId}/image	Removes the event banner image.	Organizer	None	204 No Content – image removed. 403 Forbidden – not authorized. 404 Not Found – event/image not found.

Participant Profile Picture

Part 3 also says: "Profile page where the Participant can view and update their personal information and upload their Profile Picture."

Method	Route	Description	Role	Request Body	Expected Response
POST	/api/users/profile/picture	Uploads or replaces the authenticated Participant's profile picture using Azure Blob Storage.	Participant	Multipart/form-data image file	201 Created – profile picture uploaded. 400 Bad Request – invalid file. 401 Unauthorized – not logged in. 403 Forbidden – Organizer.
DELETE	/api/users/profile/picture	Removes the authenticated	Participant	None	204 No Content – picture removed.

		Participant's profile picture.			401 Unauthorized – not logged in.